

Research Article OPEN ACCESS

A ~~real time~~ real-time reservoir inflow forecast evaluation framework

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Abstract

Recent advances in hydrologic forecasting have produced a growing number of streamflow forecast products. Hydropower operators must determine whether these products improve operations enough to justify their cost. We present a framework for comparing multiple reservoir inflow forecasts over a common period under current operating conditions. This design is useful when product-specific hindcasts cover different periods, model versions, or system configurations. Operators define the decisions and constraints to be evaluated, participate in selecting and revising metrics, and interpret results using the same observations and operating context for every product. We apply the framework to three reservoirs in the Great River Hydro system on the Connecticut River in the Northeast U.S. The commercial product labeled B generally matched the in-house forecast and outperformed product A. Under an idealized cumulative-error stress test, product B also had a lower probability of exceeding the forebay compliance band during the first 24 hours at all three dams. Its advantage persisted through the 32-hour horizon shared by all products, although exceedance probabilities were high enough that none of the forecasts could be used without operator intervention. The real-time evaluation screens products and defines the constraints for a subsequent hindcast-based value assessment using a scheduling model, operating constraints, and market prices.

1 | Introduction

~~Changing environmental regulation, diversifying and increasing water demands, and growing operational uncertainties are requiring more precise operations of reservoir~~ Changes in environmental regulation and water demand, along with growing operational uncertainty from the power grid, require more precise operation of reservoirs and hydropower systems (e.g., Lee et al. (2025) and Guo and Liu (2024) and Badr et al. (2023) and Aljoda and Jain (2021)). ~~This desire for increased precision has driven interest in improved~~ These operational demands have increased interest in streamflow forecasting tools (Yang et al. 2021 and Pagano et al. 2014 and Liu et al. 2012). ~~While among utilities and federal agencies (Yang et al. 2021 and Pagano et al. 2014 and Liu et al. 2012). With numerous commercial and in-house options are products available, water managers are faced with the challenge of evaluating whether more sophisticated forecasting tools will actually must determine whether a new product will improve their operations and enough to justify the associated financial investment cost.~~ Hydrologic forecasting and hydropower scheduling has rapidly improved in recent years due to the emergence of new data driven approaches (?) have improved through data driven models (Kratzert et al. 2019) and scheduling optimization (Zhang et al. 2024). Despite these improvements verification remains

a challenge and traditional verification approaches may not be applicable under new operational paradigms. This study seeks to provide insight into the evaluation of contemporary forecast models and guidance on how to assess their value. Hindcasts remain important because they evaluate forecast behavior over long records and can support counterfactual studies of revenue, water supply, spill, and other outcomes. Comparing several products, however, requires more than assembling their separate hindcasts. Product-specific hindcasts may cover different periods, use different model versions or meteorological inputs, or represent infrastructure and operating rules that have since changed. These differences can be mistaken for differences in forecast performance. A common real-time trial instead evaluates every product against the same observations and current operating context. This study addresses how a reservoir operator should design such a trial when comparable hindcasts are unavailable or do not represent the current system.

Pairing a forecasting tool-forecast product with the needs of a particular hydropower operator is complicated by a number of factors. The quantity and quality of data concerning meteorological (precipitation, snow pack) and hydrological (reservoir storage, soil moisture) conditions within the watershed vary locationally as do the efficacy of meteorological forecasting (Hay et al. 2023; Chen et al. 2021; Wood et al. 2016). Watershed characteristics vary as does the infrastructure used to manage basin resources (Pham et al. 2021). Operators are often subject to very different management regimes (e.g., differences in meteorological and hydrologic data, watershed properties, and infrastructure among basins (Pham et al. 2021). Operating rules range from run-of-river, rule curves (Zajac et al. 2017). Management objectives also vary according to constraints to seasonal rule curves (Zajac et al. 2017), and the relevant decisions change with forecast lead time as well as normal versus extreme event conditions. Additionally, there are decisions concerning choice of model (conceptual, physical, empirical, AI enhanced) as well as its calibration and parameterization and flow regime. An evaluation must account for these differences and connect statistical performance to decisions at a particular site.

To help provide some clarity concerning the performance of various forecasting tools under differing informational, physical, and operational conditions, a range of head-to-head evaluation exercises have been performed. HEPEX (which stands for Several programs have compared forecasts under common conditions. The Hydrologic Ensemble Prediction Experiment), which “seeks to advance the science and practice of hydrological ensemble prediction and its use in impact- and risk-based decision making,” has organized evaluations in the context of (HEPEX) has organized community experiments and testbeds¹. The, and the U.S. Bureau of Reclamation has also performed a series of forecast rodeos in a prize competition environment conducted forecast rodeos². Real-time forecast evaluations such as this (as opposed to retrospective evaluations) tend to be favored by the industry due their ability to fit in with existing. The hydropower industry often favors evaluations that fit within current operations (Krajewski et al. 2021). While such exercises provide valuable insight into the complexities and potential solutions for streamflow forecasting it doesn’t answer the question on operators’ minds—will a new forecast tool improve my operations enough to justify the investment in infrastructure, staff training and staff time during transition? These comparisons demonstrate the value of a common test period, but they do not describe how a utility should design a multi-product operating trial. The unresolved questions include which decisions and constraints to define before selecting metrics, how often operators should review results, when metrics should change, and how long the trial should continue.

Here we present an approach for assisting local water managers and hydropower utility operators in performing a context-specific evaluation of streamflow forecasting and hydropower scheduling tools. The approach calls for broad integration of the operator including experimental design, metric selection, tailored metric design, and comparative analysis of the results. We then demonstrate this approach in the context of three hydropower plants operated by Our contribution is a documented procedure for answering these design questions. Before forecasts are compared, operators define the decisions, horizons, constraints, and incumbent forecast that establish what performance means. All products are then evaluated over an overlapping period using the same observations and operating context. Operators review the results at regular meetings, identify errors that changed or could have changed a decision, and revise the metric set while preserving the reason for each change. The trial continues long enough to sample the operating regimes that matter, which is at least one annual cycle for a seasonal system. This procedure, rather than the individual verification metrics, is what distinguishes the framework from a fixed retrospective evaluation.

We demonstrate the framework at three Great River Hydro, the largest conventional hydropower provider in the Northeastern U.S. By using multiple traditional and tailored metrics as well as making the data easily accessible and comparable, this approach provides operators with the information they need to make informed decisions on forecast products with confidence. Importantly, this approach is designed to be easily replicated on other river systems plants on the Connecticut River in the northeastern US. The case study combines standard metrics with measures tied to day-ahead

¹ www.hepex.org

² <https://www.usbr.gov/research/challenges/forecastrodeo.html>

bidding and forebay compliance. Although the tailored metrics will differ among systems, the evaluation process can be adapted to other river basins and operating settings.

2 | Methods

While probabilistic and ensemble forecasts are increasingly available, deterministic forecasts such as those issued by the National Weather Service River Forecast Centers (RFC) are still widely used operationally and are adopted as a valued benchmark of this framework. In addition, RFCs remain widely used in operations in the US. All three products evaluated here are deterministic, so we focus on metrics which are feasible to compute in a real time setting as opposed to a hindcast-based evaluation. While a hindcast evaluation may be preferable statistically, hindcast evaluations require that forecasters predict (manually or with an operations model) their own behavior during hypothetical past conditions where other factors such as market prices and data curation issues may not be known. In the remainder of this section we describe the components of a real time forecast evaluation framework for reservoir inflow forecasts including visualizations to convey the evaluation results and an iterative feedback process from the operators. deterministic metrics that can be computed in real time. Section 2.5 describes how the same process can be applied to ensemble products.

Hindcasts and real-time evaluations answer different questions. A common hindcast experiment provides large samples and a broad range of hydrologic conditions. When each forecast is supplied to the same reservoir model, it can also estimate changes in revenue, water supply, spill, and other outcomes (Farkas et al. 2025). The comparison is valid only when the products use consistent inputs and represent the same infrastructure, constraints, and operating rules. Product-specific hindcasts developed at different times may not meet that requirement.

A real-time trial provides a common snapshot of current performance. It captures the forecasts as delivered, current observations and infrastructure, outages and data interruptions, operator adjustments, and the constraints that shape actual decisions. Its short record cannot represent the full range of hydrologic conditions or establish long-term economic value. The two approaches are therefore complementary: the real-time trial identifies operationally viable products and the decisions and constraints a value study must represent, while a subsequent common hindcast estimates performance and value over a longer record.

2.1 | real time Real-time inflow forecast evaluation A general real time forecast evaluation process

A multi-product real-time forecast evaluation can be described in the following steps: Collecting data, computing metrics, and visualizing results are common to any verification exercise. The contribution is the complete trial design, especially defining decisions before selecting metrics, maintaining a common comparison period, and using operator feedback to revise the evaluation.

1. **Collect real time data** Define decisions and trial scope - This may include streamflow forecasts (natural or regulated). Operators identify the decisions, forecast horizons, constraints, incumbent forecast, and operating regimes that the evaluation must represent. Products are compared over an overlapping period so that they encounter the same conditions.
2. **Collect aligned real-time data** - Archive each forecast as delivered, along with calculated inflows, scheduled and actual outflows, and other data relevant for computing evaluation-relevant metrics such as forecast issue times and past forecasts issue times, observations, outages, and other data needed to interpret performance.
3. **Compute evaluation initial metrics** - At the start of an evaluation, it may only be possible to compute a limited number of generic evaluation metrics but as the available data grows it may become possible to compute more reliable metrics as well as metrics tailored to the specific system. Begin with standard metrics that can be calculated from the available sample. Compute each product on common observations whenever coverage allows.
4. **Visualization Visualize results** - Developing easy to interpret visuals is critical to the success of an evaluation. Visuals should be concise, clear, and require a limited amount of explanation, or come with built-in documentation. Develop clear figures that allow operators to compare products and identify differences that matter to operations.
5. **Gather feedback** - At regular meetings, ask operators which recent forecast errors affected or could have affected a decision, and which errors did not matter operationally.
6. **Solicit feedback Update metrics** - Collecting feedback on an ongoing basis from operators is critical to the success of a real time forecast evaluation. This process can be used to develop tailored metrics that are specific to the system or

reservoir being evaluated. Add metrics tied to the identified decisions and constraints, and modify or remove metrics that proved uninformative. Record why each change was made.

7. **Iterate** - The above steps should be repeated at a regular interval (Repeat the review and update cycle, monthly or weekly) throughout the entire forecast evaluation period, ideally, while accumulating a sample that covers the operating regimes of interest. For a seasonal system, the trial should span at least one year in duration annual cycle.

The iterative nature of this evaluation process with direct operator feedback is important to its success. As more data becomes available so does an understanding of the forecast performance which may help to develop additional tailored metrics. The following sections detail some additional considerations. mechanics of steps 5 and 6 determine what the evaluation produces, so an example of how feedback was collected and acted on in our case helps clarify this point. Feedback was collected in recurring meetings with the operators, held monthly for most of the evaluation. Each meeting reviewed the current calculated metrics against recent operating experience, and operators were asked to identify which recent forecast errors mattered operationally, and which did not. The answers frequently diverged from what the generic metrics indicated, especially during the high-flow season. Operators discounted large errors during high-flow spill conditions, when they had little discretion over releases and forecast quality was operationally irrelevant, while treating moderate errors with the potential to impact the forebay compliance band as serious. These ongoing discussions motivated the tailored metrics in Section 3.6: the forebay exceedance metrics were designed in direct response to operators stating that compliance risk, rather than average error, determined whether they trusted a forecast.

The iteration in step 7 occurs over two timescales. In this case, the set of metrics stabilized within about six months, while the evaluation continued to collect data under a wider range of conditions. For applications with strong seasonality, including this one, we recommend at least one annual cycle because product rankings can vary by season (Section 3.5). Operators should expect the metrics to stabilize before the evaluation period ends. Conclusions from a shorter evaluation remain conditional on the seasons and events observed.

2.2 | Benchmark forecasts

It is critical to include benchmark forecasts in the evaluation to provide forecast evaluators with context for how well a specific forecast performs over the benchmarks. Two commonly used benchmarks are the Benchmark forecasts put product performance in context by showing how much improvement a product provides over a known alternative. A “perfect” forecasts, which use forecast substitutes observed or calculated inflows in place of a forecast to get a sense of the best possible forecast, and “persistence” forecasts which use the last observed value carried out for all future timesteps to get a sense of the arguably worst possible forecast inflow for a forecast and represents the upper reference. A persistence forecast carries the most recent observation through the forecast horizon. Persistence is a competitive benchmark for short term operations, from hours to 2-day horizons. We note that other simple climatologically driven forecasts could be used in place of the persistence forecast when looking at longer horizons like monthly and seasons. Other benchmarks could include publicly available forecast products or any other forecast that is currently being used at the reservoir simple but can be competitive from hourly lead times through about two days, making it an appropriate reference for day-ahead operations. It degrades quickly beyond that horizon, where climatology is usually a more suitable lower reference. Evaluators should match the benchmark to the decision horizon rather than apply one reference at all lead times. Other useful benchmarks include public forecast products and the forecast currently used by the reservoir operator.

Benchmarks are also useful when comparing results among locations. NSE and KGE depend on the distribution of observations at each location, so their raw values are not directly comparable when inflow variability differs (Knoben et al. 2019). NSE, for example, measures improvement over the observed mean. The same absolute forecast error can produce different NSE values at two sites because the observed variances differ. A forecast can therefore have a higher NSE at a more variable site even when its absolute errors are larger. We report RMSE and reference forecasts alongside NSE and KGE so that the results show both absolute error and performance relative to alternatives available to the operator.

2.3 | Smoothing calculated inflows

Most available reservoir inflow data is not directly measured, instead it is computed based on measured Reservoir inflow is often calculated from changes in forebay elevation and a known an elevation-volume relationship. These calculations are often noisy due to short term fluctuations in forebay elevations from rather than measured directly. Short-term changes caused by wind, turbine and spillway release changes, and other such as reservoir seiches or waves propagating from upstream. Inflow computed in this way can often contain negative values and other non-physical fluctuations. In this situation we recommend the inflow data be smoothed prior to comparing with any forecasted inflow. We recommend

a triangular smoother that applies moving averages using a triangular weight function. This is a simple option which produces reasonable results but many more options releases, seiches, and waves from upstream can introduce noise and nonphysical values into the calculated inflow. Smoothing may therefore be appropriate before calculated and forecast inflow are compared. We used two successive moving averages, which produce a triangular weighting function, but other filters are available (Elshorbagy et al. 2002). Figure 3 shows an example of hourly computed inflow data which has been smoothed using a triangular smoother. The smoothing window should correspond to the timescale of the decision. A window that is too short will retain noise, while one that is too long may remove variability that the forecast should reproduce. Because smoothing changes the quantity being evaluated, the filter should be reported and the sensitivity of the results to its width should be tested.

The top figure shows the unsmoothed inflow and bottom figure shows the smoothed inflow used in the evaluation. The black lines show the calculated inflow based on forebay observations and dam outflow and the gray lines show the forecasts for reference.

2.4 | Traditional inflow evaluation metrics

Traditional deterministic forecast evaluation metrics that are particularly relevant to inflow forecasting include Deterministic inflow forecasts can be evaluated using measures of average behavior error, correlation, and bias. Additional metrics include higher order moments of the forecast distributions, categorical forecast Higher moments, categorical performance, and conditional statistics (St-Aubin and Agard 2022 and Hewamalage et al. 2023). This section is not intended to be a comprehensive may also be useful (St-Aubin and Agard 2022 and Hewamalage et al. 2023). We do not attempt a complete review of deterministic forecast evaluation metrics. Therefore, we recommend some simple metrics which can be augmented as needed for particular applications. These metrics should be computed for a variety of lead times as well as seasonally to fully assess forecast performance across a range of conditions metrics. Instead, we use a small set that describes several aspects of performance and can be expanded for a particular application. No single metric fully describes forecast quality. The metrics should also be calculated by lead time and season because values aggregated across the full record can hide changes that matter to operations.

Average hydrologic forecast behavior is typically measured with an error metric such as root mean squared error (RMSE), Nash-Sutcliffe Efficiency (NSE), or King-Gupta (Nash and Sutcliffe 1970), or Kling-Gupta Efficiency (KGE) (refs) (Gupta et al. 2009). We recommend including KGE because it is not skewed by large values where decomposes forecast performance into correlation, a variability ratio, and a bias ratio. A low score can therefore be associated with a specific deficiency rather than registered only as a large aggregate error. NSE and RMSE are due to the use of a squared error term aggregate squared error into a single number, which makes them more sensitive to the largest errors and less diagnostic about their source. The metrics answer different questions, so we report them together rather than use one as a complete measure of forecast quality. The KGE is given by

$$KGE = 1 - \sqrt{(r - 1)^2 + \left(\frac{\sigma_f}{\sigma_o} - 1\right)^2 + \left(\frac{\mu_f}{\mu_o} - 1\right)^2} \quad (1)$$

where r is the linear correlation coefficient, σ_f is the standard deviation of the forecast, and σ_o is the standard deviation of the observations, are the forecast and observed standard deviations, and μ_f is the mean of the forecast, and μ_o is the mean of the observations. Values of KGE that are greater than -0.41 are better than the mean of the observations, but larger values are preferred up to are the forecast and observed means. KGE values greater than -0.41 improve on the observed mean as a predictor (Knoben et al. 2019), and the maximum value of is 1 for a perfect forecast. We note that NSE and KGE have different properties and so Because the corresponding reference value for NSE is 0, the two efficiency metrics should not be used interchangeably (Knoben et al. 2019); interpreted on the same scale. We report both and use RMSE, bias, correlation, and benchmark forecasts to interpret their values.

The bias error, or residual, of a single forecast point is the difference between observed and forecasted values but this may be expressed as a percentage the forecast and the observation. Bias is the systematic component of error, the tendency of a forecast to be too high or too low on average, and is conventionally expressed for a set of forecasts by averaging across all bias values as a percentage of total observed volume

$$PBIAS = 100 [\%] \times \frac{\sum(f_i - o_i)}{\sum(o_i)} \quad (2)$$

where PBIAS measures the tendency of a forecast (f) to be greater than (positive values) or less than (negative values) than observations (o) positive values indicate forecasts greater than observations and negative values indicate forecasts less than observations.

Correlation measures the strength of a linear relationship Linear association between observations and forecasts is computed measured using the Pearson correlation coefficient

$$r = \frac{\sum(o_i - \mu_o)(f_i - \mu_f)}{\sqrt{\sum(o_i - \mu_o)^2} \sqrt{\sum(f_i - \mu_f)^2}} \quad (3)$$

where r can range from -1 to 1 but for forecast evaluation, negative values would indicate poor forecast performance. Metrics should be computed on a common sample when products cover different periods. Skill scores relative to a reference forecast are also useful, but the evaluated and reference forecasts must use the same observations. Otherwise differences in sample variance can be mistaken for differences in skill.

2.5 | Extension to probabilistic and ensemble forecasts

The process described in this section applies to both deterministic and probabilistic forecasts; only the metrics and visualizations change. Probabilistic products can describe uncertainty around compliance or market decisions rather than provide only a single expected value. Where ensemble or probabilistic products are available, the continuous ranked probability score (CRPS) can summarize the accuracy of the full predictive distribution. CRPS reduces to mean absolute error for a deterministic forecast, which permits comparison on a common scale. Rank histograms can indicate whether an ensemble is biased or has too little or too much spread, while reliability diagrams compare forecast probabilities with observed frequencies. The choice among these metrics should follow the same process used for deterministic forecasts and reflect the decisions being evaluated.

3 | Case Study: Great River Hydro

Great River Hydro (GRH) is New England's largest conventional hydropower generator. As a private, independent power producer, GRH owns and operates 589 megawatts of nominal hydropower capacity installed nameplate capacity across 13 hydropower plants on the Connecticut and Deerfield Rivers in ISO New England. In this evaluation we focused on the inflow to three main hydropower producing dams in the system We evaluated inflow forecasts for three mainstem dams, Wilder, Bellows Falls, and Vernon (Figure 1) as well as three main, and for three tributaries, the Ottauquechee, Sugar, and White Rivers.

The three dams lie in series on the main stem of the Connecticut River, with Wilder upstream, followed by Bellows Falls and Vernon. They are operated jointly, and releases from each dam become inflow to the next. The three tributaries listed above provide the main local inflows between Wilder and Bellows Falls. We selected these sites from the larger set shown in Figure 1 because they are the largest generators in the system and receive most of the operators' scheduling attention. The projects are also part of an ongoing relicensing process through the Federal Energy Regulatory Commission (FERC) that includes new limits on forebay fluctuations. The remaining facilities in Figure 1 are smaller and are not covered by the same license conditions.

Basin hydrology is snowmelt influenced, with a spring freshet typically spanning March through May, convective rainfall in summer, and rain and rain-on-snow driven events in winter. Several tributaries have upstream flood control regulation by the U.S. Army Corps of Engineers, which contributes to inflow variability that is not directly weather driven.

TABLE 1 | Physical characteristics of the six evaluation points. Drainage areas are from the USGS National Water Information System for the reference gage listed. Mainstem areas are nested, so the area above Vernon includes those above Bellows Falls and Wilder.

Location	Type	USGS reference gage	Area (mi ²)	Area (km ²)
Wilder	Mainstem dam	01144500 West Lebanon, NH	4,092	10,598
Bellows Falls	Mainstem dam	01154500 North Walpole, NH	5,493	14,227
Vernon	Mainstem dam	01156500 Vernon, VT	6,266	16,229
White River	Tributary	01144000 West Hartford, VT	690	1,787
Sugar River	Tributary	01152500 West Claremont, NH	269	697
Ottawaquechee River	Tributary	01151500 North Hartland, VT	221	572

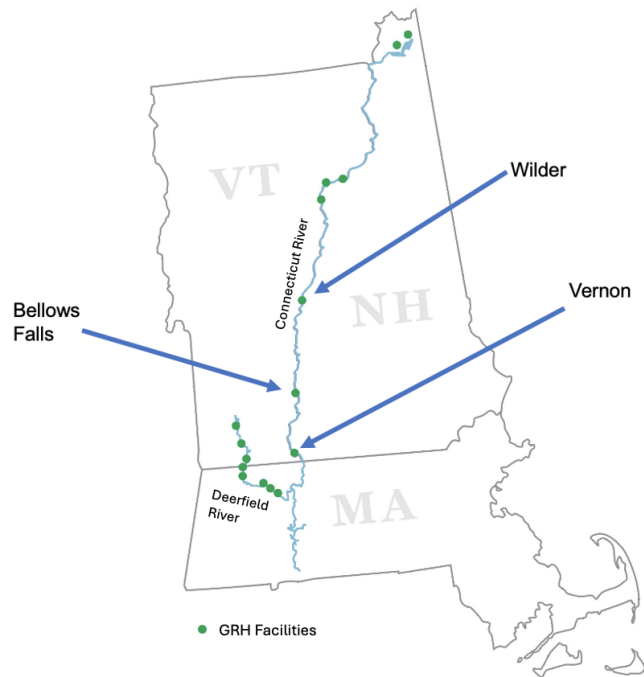


FIGURE 1 | The Great River Hydro system in the Northeast U.S. with the locations of the three major dams in this evaluation.

3.1 | Operating context

The three dams have limited storage relative to their inflow and are operated as run-of-river facilities. Their daily cycle consists of an upward ramp, a period of flexible generation, a downward ramp, and a refill period. Generation is bid into the ISO New England day-ahead market, so the day-ahead inflow volume is the quantity that matters most to the operators, since it sets what they can bid into the market. The binding operational constraint comes from the FERC relicensing of the three lower Connecticut River dams, which imposed a ± 0.5 foot (≈ 15 cm) limit on forebay fluctuation, with penalties for exceedance. The central operating problem is therefore committing generation based on an uncertain inflow forecast without violating the forebay limits. This is what motivated the cumulative-error and threshold-exceedance metrics in Section 3.6.

At the time of the evaluation the operators used an in-house forecast built from National Weather Service River Forecast Center forecasts, routed to the dam locations and adjusted using operator judgment. This is the incumbent product against which the two commercial forecasts were evaluated, and it is labeled GRH in the results.

Setting up a ~~real-time~~ real-time evaluation required the development of an automated data platform that enabled sharing of ~~real-time~~ real-time system data as well as the forecasts. This was necessary to share certain data with external parties for the purposes of forecast evaluation. While such a system may not be necessary for evaluations conducted internally by a single organization, some form of data access (API, database, etc.) is required for a ~~real-time~~ real-time forecast evaluation. Such systems incur an up-front cost that should be considered when deciding on an evaluation approach. In addition, ~~real time~~ real-time evaluations come with all of the challenges of any operational system, such as system uptime, latency, and maintenance. These factors may help decide if a forecast is viable before any evaluation is done; a forecast that cannot be produced operationally should not be considered in a real-time evaluation.

We ~~ran the real-time~~ conducted the real-time evaluation for almost ~~2 years~~ from July two years, from June 2023 to May through April 2025, with regular feedback from the operators. We evaluated two commercial forecast products ~~which are anonymized and labeled~~, anonymized as A and B. ~~In addition we evaluated~~, and the GRH in-house forecast ~~which is a combination of~~, which combines RFC forecasts, routing, and operator judgment. During the ~~initial few months of the evaluation~~ first few months, we focused on ~~generic metrics~~ general measures of forecast performance. As the ~~quantity of data increased we transitioned to seasonally computed metrics and tailored metrics developed in collaboration~~ archive grew, we added seasonal summaries and metrics developed with the system operators.

~~While most weather-driven inflow forecasts are available hourly for the next 7-10 days, The products cover different portions of this period, which limits direct comparison. Forecast B was available for the volumetric forecast for the next day is the most important for GRH. This largely sets what they can bid into the day ahead market. Table 2) show the overall forecast evaluation metrics for the two commercial forecasts (A full period, providing 657 day-ahead forecasts at~~

each dam. Forecast A was discontinued in November 2024, and the in-house forecast archive also ends in November 2024, providing 539 and 528 day-ahead forecasts, respectively. The three products overlap on 510 days at each dam, from 3 June 2023 through 20 November 2024. We report metrics computed from each product's full record, so the samples are not identical. We repeated the calculations using the 510 common days. The product rankings were unchanged at all three dams, and individual NSE values changed by no more than 0.03. We therefore retained the full record for each product. Partial coverage of this kind is a practical limitation of real-time evaluations.

3.2 | Forecast products

Anonymization of the commercial products limits reproducibility, so we describe each to the extent our agreements allow. Forecast A is produced by a consulting firm using a conceptual rainfall-runoff model with hydraulic routing through the Connecticut River main stem, driven by a commercial numerical weather prediction forecast, and issued once daily at an hourly time step out to roughly 8 days. Forecast B is produced by a technology firm using a machine learning model trained on gauge and remotely sensed catchment observations, issued twice daily at an hourly time step out to 10 days. The GRH in-house forecast combines RFC forecasts with routing and operator judgment and extends only through the day-ahead period, roughly 32 hours, because that is the horizon it is used for. Neither vendor released model internals or calibration and validation diagnostics, and we did not have access to them. A utility evaluating commercial forecast products generally cannot inspect them and must judge them on operational performance, which is part of the motivation for this framework. A full procurement evaluation should also weigh the operational cost, data dependencies, and ease of integration of each product alongside its skill, since these differ substantially between the two commercial products evaluated here.

3.3 | Hydrology of the evaluation period

The evaluation period included the July 2023 Vermont flood. This event produced the largest calculated inflows of the evaluation at Vernon (83, B) 100 cfs, 2, the GRH 353 m³/s) and Bellows Falls (67,000 cfs). A December 2023 rain-on-snow event produced the largest inflow at Wilder (27,600 cfs) and the second largest at Vernon and Bellows Falls. The evaluation period also included spring freshets of 2024 and 2025 and dry summer conditions in both 2023 and 2024. The period therefore sampled a reasonable range of the conditions relevant to GRH operations. It did not, however, sample the full distribution of hydrologic conditions. Notably, the July 2023 event produced one of the ten largest flows ever recorded on the main stem of the Connecticut River. The evaluation also did not include any severe multi-year drought. Any real-time evaluation is bounded in this way, and the conclusions below should be taken as conditional on the observed period. Most weather driven inflow forecasts extend 7–10 days, but GRH relies most on next-day volume because it informs the day-ahead market bid. Table 2 compares the two commercial products, the in-house forecast (GRH), and the benchmarks perfect and persistence forecasts for the one-day-ahead forecasts. Overall we found that for this time period, GRH's benchmarks at this horizon. Over the evaluation period, forecast B and the in-house forecast was roughly similar to forecast B but better than forecast A overall. This overall tabular view is a good way to get quick comparisons between forecasts but can obfuscate certain forecast behaviors, especially seasonal. performed similarly, and both outperformed A. The lead-time and monthly results below show variation that is hidden by the aggregate values. Table 2 also shows why cross-location comparisons need more than one metric. Forecast A has its highest NSE at Vernon (0.81) and its lowest at Bellows Falls (0.55), yet its RMSE is 5,097 cfs at Vernon, 5,904 cfs at Bellows Falls, and 2,278 cfs at Wilder. NSE is normalized by observed variance, so a higher value at one site does not imply a smaller error than at another. The persistence benchmark also changes sharply among sites, from an NSE of −0.81 at Wilder to 0.80 at Vernon. We therefore use the table to compare products within a location, while RMSE and the benchmark rows provide context across locations. Metrics computed on the 510 days common to all products preserve the within-location ranking. Figure 2 shows the performance of all the available forecasts and benchmarks out to a 10-day forecast and benchmark performance through a 10-day lead time. The GRH in-house forecast is only used to develop bids for the day-ahead market so it is used for day-ahead market bids and does not extend the full 10-day through the full period. At Wilder and Vernon all, the three forecasts perform roughly similar with 1–3 days of lead time with the similarly over the first 1–3 days, after which A and B forecasts diverging after that period. At Bellows Falls the three forecasts are distinct in terms of quality with the B showing the best performance diverge. The differences are larger at Bellows Falls, where B performs better at most lead times, followed by the GRH in-house forecast and then forecast A. Notably, after three days of A. Both commercial forecasts develop an increasingly negative bias with lead time, all forecasts exhibited some amount of bias at the downstream forecast locations which may be due to errors propagating from upstream, and this bias increases downstream, consistent with errors propagating through the cascade. At Bellows Falls, for example, the PBIAS for A

TABLE 2 | Overall Day-ahead forecast evaluation metrics for the real-time evaluation conducted from July-June 2023 to May-2025 for the two commercial forecasts (A and B) are commercial products, the GRH is the in-house forecast (GRH), and the benchmarks perfect and persistence are benchmarks. The products cover different portions of the period; n is the number of forecasts evaluated. Values should be compared primarily within a location because NSE and KGE depend on the local observation distribution.

Location	Forecast	Location- n	NSE	KGE	PBIAS (%)	rRMSE (cfs)	r
Wilder	A	Wilder-539	0.72	0.83	-8.00-8.0	2,278	0.88
	B	Wilder-657	0.83	0.88	0.10 0.1	1,786	0.93
	GRH	Wilder-528	0.88	0.89	-9.90-9.9	1,476	0.95
	perfect	Wilder-657	1.00	1.00	0.00 0.0	0	1.00
	persistence	Wilder-657	-0.81-0.81	0.02	-75.90-75.9	5,866	0.75
Bellows Falls	A	Bellows-539	0.55	0.77	0.40 0.4	5,904	0.77
	B	Bellows-657	0.91	0.95	0.60 0.6	2,615	0.95
	GRH	Bellows-528	0.85	0.88	-10.40-10.4	3,309	0.93
	perfect	Bellows-657	1.00	1.00	0.00 0.0	0	1.00
	persistence	Bellows-657	0.39	0.35	-39.20-39.2	6,831	0.87
Vernon	A	Vernon-539	0.81	0.83	-11.30-11.3	5,097	0.91
	B	Vernon-657	0.89	0.94	1.70 1.7	3,872	0.94
	GRH	Vernon-528	0.89	0.88	-11.00-11.0	3,764	0.95
	perfect	Vernon-657	1.00	1.00	0.00 0.0	0	1.00
	persistence	Vernon-657	0.80	0.69	-18.10-18.1	5,167	0.93

changes from 0.4% at the day-ahead horizon to -19.1% by day 7. The corresponding values for B are -1.3% and -7.8%. Because releases from each dam become inflow to the next, forecast errors can propagate to the next downstream project and accumulate with lead time.

Figure 4 shows metrics by month for the entire evaluation period which reveals some trends in forecast quality in certain seasons. Namely forecasts tend to perform worse in the winter which is the wet season and worse in the summer which is the dry season. Forecasts tended to perform well during the snow melt season (April-July

3.4 | Smoothing GRH calculated inflows

Figure 3 shows calculated inflow before and after smoothing. We applied two successive 24-hour moving averages to the hourly calculated inflow. Their convolution yields a triangular kernel spanning 47 hourly observations; “24-hour” refers to the width of each component moving average. We selected this width to match the day-ahead volumetric decision and remove fluctuations at shorter timescales. Because the window directly affects the metrics, we repeated the day-ahead evaluation with component widths of 0, 6, 12, and 24 hours (Table 3). The natural hydrology of the system is driven by different mechanisms in different seasons metrics improve slightly as the width increases, as expected when observation noise is removed. The largest change is 0.05 NSE. The ranking of the products is unchanged at every width and location. At Bellows Falls, for example, snow vs rain and baseflow vs direct runoff, which may explain the varying model performance by season. Although forecast B tends to perform better overall, there are cases in which the B leads A by 0.39 NSE without smoothing and by 0.37 with the 24-hour component width. The conclusions therefore do not depend on the selected window. Evaluators should still report the filter and confirm that their results are insensitive to its width because excessive smoothing could favor forecasts that underrepresent inflow variability.

TABLE 3 | Sensitivity of day-ahead NSE to the component moving-average width used to smooth calculated inflow. Each nonzero width is applied twice, producing a triangular kernel. The 24-hour column corresponds to Table 2. Product rankings are unchanged.

Location	Forecast	0 h	6 h	12 h	24 h
Wilder	A forecast is better, highlighting the difficulty of selecting a single deterministic forecast-	0.70	0.72	0.72	0.72
	B	0.82	0.83	0.83	0.83
	GRH	0.87	0.87	0.88	0.88
Bellows Falls	A	0.50	0.51	0.52	0.55
	B	0.90	0.90	0.90	0.91
	GRH	0.84	0.83	0.84	0.85
Vernon	A	0.79	0.79	0.79	0.81
	B	0.88	0.88	0.88	0.89
	GRH	0.88	0.88	0.88	0.89

3.5 | Seasonal performance

Figure 4 shows monthly variation in forecast quality. All three products have their highest skill during the spring snowmelt period, roughly March through May. The commercial products have their lowest skill in late summer and fall. Forecast A also degrades in January and February, when its day-ahead KGE pooled across the three dams falls to 0.45 and -0.67, respectively; the February value is below the plotted range. Forecast performance is lower when inflow depends on rain, rain-on-snow, or convective storms, all of which depend on uncertain precipitation forecasts. Snowmelt inflow is more predictable because the snowpack can be observed before it melts and temperature forecasts are generally more reliable than precipitation forecasts. Forecast A outperforms B in April and May, so no product leads in every month.

3.6 | Tailored metrics

In addition to the ~~generic forecast evaluation metric~~ general evaluation metrics described above, we developed several ~~tailored metrics specifically relevant to metrics for~~ the GRH system and its operations. ~~The first focused on day-ahead forecast performance as shown in the previous section. This was motivated by the day-ahead market that GRH bids into, as such they focus their attention heavily on the 1-2 day-ahead period where the forecasts matter the most. Operationally, they adjust their flows after this point, so no forecast is used after the 2-day mark.~~ Each arose from the feedback process described in steps 5 and 6 of Section 2.1. Table 4 traces each metric to the operator concern that prompted it. The first measures day-ahead performance because GRH uses the next day's expected inflow to prepare its day-ahead market bid. Operators adjust flows daily, so they concentrate on the first two days and do not use a forecast beyond that horizon. ~~During the Federal Energy Regulatory Commission (FERC) relicensing for the three main dams on the lower Connecticut river a strict limit on forebay fluctuations was imposed. GRH was given a +/-~~

TABLE 4 | Tailored metrics developed during the evaluation, the operator concern that prompted each, and the approximate point in the evaluation at which it was added. Metric development converged within roughly the first six months.

Metric	Operator concern	Added
Day-ahead volumetric skill	Day-ahead market bids are set from next-day inflow volume, not hourly shape	Months 1–2
Seasonal metric breakdown	Products were suspected to behave differently in snowmelt and rain driven regimes	Months 3–4
Cumulative forebay error	Average error does not indicate whether the forebay band would be breached	Months 4–6
Forebay exceedance probability	Operators needed to know how quickly cumulative forecast error could exhaust the forebay band	Months 5–6

The FERC relicensing described in Section 3.1 imposed a ± 0.5 foot ~~operating range after which penalties are incurred. Given GRH's reliance on the inflow forecast, a metric was developed that captures the chance of the forebay limit being exceeded if the inflow forecast is used directly~~ (≈ 15 cm) forebay operating band. We calculated cumulative hourly forecast error within each forecast issue and converted the accumulated volume to an equivalent forebay elevation change using an approximate storage scale of 30,000 cfs-hours per foot. Figure 5 shows ~~where the forebay would be after a certain amount of time if the forecast was used directly to set a pass inflow condition (i.e., inflow=outflow).~~ The error bars extend to the 2.5th and 97.5th percentile of all forecasts. We can see that rarely would a forecast cause the forebay to exceed the 0.5 foot limit during the first day of operations, after which the operators can adjust. ~~the median and middle 95% of the resulting trajectories. This is an idealized stress test: it assumes forecast inflow is passed directly as outflow from a neutral forebay position. GRH does not operate this way. Operators schedule generation to market bids, depart from inflow within the day, and adjust as new information arrives. The metric is therefore neither a prediction nor a bound on observed compliance rates. Estimating realized risk requires a scheduling model and operator review.~~

Figure 6 (top) shows the corresponding probability of exceeding the 0.5 foot limit given a neutral forebay starting position on day 0. Results suggest that chances of exceedance are less than 10% for all forecasts for all lead-times in the 1-3 day range. GRH was also interested in seeing a more realistic situation involving a non-optimal forebay starting position on the second day, what was the probability of exceeding the 0.5 foot range, which is shown in Figure 6 (bottom). We can see that the probabilities are much higher, up to 25% in some cases for the A forecast. This information provides a detailed view of the forecast performance that GRH can use to assess forecast quality. ~~leaving the forebay band. Over the first~~

24 hours of each product's horizon, forecast B has lower exceedance probability than A at all three dams: 31.0% versus 56.6% at Wilder, 47.7% versus 67.9% at Bellows Falls, and 53.8% versus 71.1% at Vernon. The in-house forecast, whose horizon extends only 32 hours past issue, gives 53.9%, 67.6%, and 73.3% respectively, placing it close to A rather than to B. By 32 hours the ordering is unchanged: B reaches 42.8–65.5% across the three dams while A reaches 61.2–82.0% and the in-house forecast 60.2–85.4%. Thus B holds a consistent advantage across the whole window the three products share, but no product can be passed through without intervention: by the end of the first day every one of them leaves the band in roughly a third to three-quarters of issues, and the exceedance probability rises monotonically with each additional hour accumulated. This distinction was not apparent from the aggregate metrics alone.

4 | Discussion

In this paper we present a general framework that can be used in real time (or near real time) for reservoir inflow forecast evaluation. The evaluation can include forecast verification and validation but also include other qualitative or specialized metrics of forecast quality. The framework calls for an iterative approach which involves direct operator feedback in the process. We also presented a successful application of this evaluation framework. We present a framework for comparing multiple reservoir inflow forecasts over the same period and under current operating conditions. Its contribution is the design of the operating trial: define the decisions and constraints before selecting metrics, align the product records, review results with operators at regular intervals, preserve why metrics change, and continue long enough to sample the relevant operating regimes. We applied this process to three reservoirs in the Great River Hydro system on the Connecticut River in the Northeast U. S. While there is no single evaluation approach that can apply to all forecasts, we present a range of options of both traditional and tailored metrics that will be suitable for many situations. The goal of such a framework is to provide operators with information necessary for evaluating how much a particular forecast product will improve their operations and ultimately inform ongoing investment decisions (i.e. purchase external forecast products or improve existing internal forecast). Product B generally matched the in-house forecast and outperformed product A. The forebay stress test expressed this difference in operational units and showed how risk accumulated over the shared forecast horizon. These results screen forecast products and define requirements for a later value analysis, but they do not estimate revenue or observed compliance outcomes. In our real time evaluation of the GRH inflow forecasts there was one commercial forecast that stood out as having better performance overall. However, it was not always the best.

4.1 | Comparison with retrospective forecast evaluation

Hindcasts and real-time evaluations are complementary when they are designed as two parts of one comparison. A common hindcast can evaluate the same products across decades of hydrologic variability and, when coupled to a reservoir model, estimate effects on revenue, water supply, spill, and other outcomes. Separate vendor hindcasts are not necessarily comparable if they use different periods, model versions, meteorological inputs, infrastructure, or operating rules. A common experiment must hold these elements constant. The real-time trial supplies information that a hindcast usually omits. It tests products as they are delivered against the same current observations, and it records data interruptions, operator adjustments, current constraints, and the decisions that forecast errors affect. The limited period cannot establish long-term value or sample all hydrologic regimes. Its role is to determine which products are operationally viable and what a subsequent hindcast must represent. If comparable hindcasts already exist, the two evaluations can proceed in parallel. Otherwise, the real-time trial should precede the value study so that the hindcast uses current constraints and decision rules rather than an outdated system representation.

4.2 | From forecast quality to forecast value

The metrics presented here measure forecast quality rather than forecast value. Quality describes the relation between forecasts and observations. Value also depends on the decision, its consequences, and the decision maker's tolerance for risk. A more accurate forecast has no additional value if it does not change an action. Classical cost-loss analysis links forecast use to the utility of a decision (Thompson and Brier 1955). The relative utility value framework extends this approach to binary, multicategory, and continuous decisions and can represent risk aversion (Laugesen et al. 2023). Farkas et al. (2025) review related methods. The tailored metrics are a step toward value because they express forecast error in units related to a decision. The forebay result does not demonstrate that compliance violations were avoided. Instead, it shows that a valuation cannot

assume unrestricted use of the inflow forecasts: under the pass-inflow stress test, cumulative error frequently exhausts the operating band and requires operator intervention. A subsequent hindcast-based value study should therefore admit only schedules that satisfy the forebay constraint and should represent how operators respond as the band is approached. At GRH, the real-time trial can first screen products and define these constraints. The selected forecasts can then drive a common scheduling and dispatch model containing reservoir constraints, unit performance, day-ahead and real-time prices, and the current operating strategy. Benefits should be measured relative to the forecast and decisions now in use and tested under different operating strategies and attitudes toward risk.

4.3 | Limitations and next steps

Forecast B performed better overall, but it did not lead in every month and every lead time. This underscores or at every forecast horizon. This variation illustrates the difficulty of selecting a single best deterministic forecast. Moving to a probabilistic forecast may alleviate this somewhat but operators still need to bid in a fixed amount into energy markets so overall improvements in deterministic forecast skill are valuable. In this study we did not address how much one deterministic forecast for all conditions. Product selection should account for the decisions, seasons, and horizons that matter most to the operator. Probabilistic forecasts can describe uncertainty around these choices, although operators must still submit fixed market bids.

The results also show why forecast performance should not be compared among locations using raw efficiency metrics alone. NSE and KGE depend on the distribution of observations, so a high score at one site does not necessarily imply smaller errors than a lower score at another site. RMSE and reference forecasts provide additional context. Comparisons among products should also use the same dates when their coverage differs. Here, restricting the analysis to the common period did not change the product rankings.

We did not estimate how improvements in inflow forecasts will translate into improvements in hydropower operations and increased revenue. An important next step in this evaluation framework is to couple this evaluation with hydropower scheduling models to quantify increases in revenue from improved inflow forecasts. While this seems a straight forward next steps, many hydropower scheduling models work with ranges of operations that match the utility's bidding strategy and local market opportunities. This complexity implies that evaluating the opportunities for harnessing higher forecast accuracy may also requires updating scheduling models and exploring more monetized operations affect hydropower operations or revenue. That analysis would require coupling the archived forecasts to a hydropower scheduling model. A model configured to reproduce the current bidding strategy may understate the benefit of a more accurate forecast because that strategy was developed around the incumbent product. Estimating value may therefore require changes to the scheduling model and operating strategy as well as the forecast input.

Real time forecast evaluations are inherently limited by the amount of time they are conducted. Ideally a forecast model's performance would be assessed over the full range of hydrologic conditions spanning multiple decades. A hindcast evaluation may account for the full The length of the available record also limits the evaluation. The period included a regional flood and a rain-on-snow event (Section 3.3), but it did not include a severe multi-year drought. We could not assess forecast performance under sustained water-limited conditions. A multi-decade hindcast would include a wider range of hydrologic variability but this requires an operations model or considerable effort on the part conditions and provide a more complete evaluation of forecast quality.

The choice between real-time and retrospective evaluation depends on the needs of the system operators to re-operate the system for every hindcast. The decision to pursue a real time or hindcast evaluation must be made by the system operators depending on their specific system needs and resource availability and the available resources. We recommend conducting both when feasible. For a system with strong seasonal variation, a real-time evaluation should span at least one year. A shorter period samples only part of the annual cycle, and its conclusions remain conditional on the seasons and events observed.

The iterative nature of this framework is perhaps its most valuable component. Having direct operator involvement and feedback helps to provide value to Two other limitations affect the interpretation of the results. First, the reference inflows are calculated from forebay elevation changes and dam outflow. The observations therefore depend on reservoir operations and are not independent in the same way as a stream gauge record. Second, the three products cover different portions of the evaluation period (Section 3.1), so the metrics in Table 2 use overlapping but nonidentical samples. Restricting the analysis to common dates does not change the ranking. We report the full records to retain the available data.

The iterative component of the operators. The development of tailored metrics make the results relevant to a particular system and helps with operator buy-in. Furthermore, both generic and tailored metrics can ultimately be extended to assess the impacts of inflow forecasts on hydropower scheduling. Particularly under dynamic electricity market

conditions, evaluations of inflow forecasts can serve as key indicators for understanding how forecasting errors propagate through the system, influencing compliance with operational requirements, variations in actual power generation, and potential revenue for system operators. The analysis of inflow forecast performance, conducted in coordination with system operators, represents an important foundational step toward achieving these broader applications. In presenting this framework we hope to provide a template for further applications and unbiased evaluation of the myriad of inflow forecasts available to hydropower system operators. Our framework allows the evaluation to change as operators identify which forecast errors affect their decisions. Standard metrics provide a consistent basis for comparison, while tailored metrics connect forecast errors to the constraints of a particular system. Together, these metrics provide the forecast archive and initial assessment needed for a later scheduling and value analysis. The same process can be applied elsewhere, but the relevant decisions and tailored metrics must be defined with the operators of each system.

DATA AND CODE

Code for the evaluation is freely available³. Sample data can be accessed from the data repository on Zenodo⁴.

~~Note that a pre-print non-peer-reviewed draft~~ A preprint version of the manuscript ~~has been uploaded to~~ is available at <https://eartharxiv.org/repository/view/10530/>.

AUTHOR CONTRIBUTIONS

Cameron Bracken - Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - Original Draft Preparation. Youngjun Son - Conceptualization, Data Curation, Formal Analysis, Investigation, Software, Validation, Visualization, Writing - Original Draft Preparation. Vince Tidwell - Conceptualization, Formal Analysis, Investigation, Methodology, Project Administration, Supervision, Writing - Original Draft Preparation. Nathalie Voisin - Conceptualization, Methodology, Project Administration, Supervision, Funding Acquisition, Writing - Acquisition, Writing - Original Draft.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

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³ <https://github.com/HydroWIREs-PNNL/inflow-forecast-evaluation>

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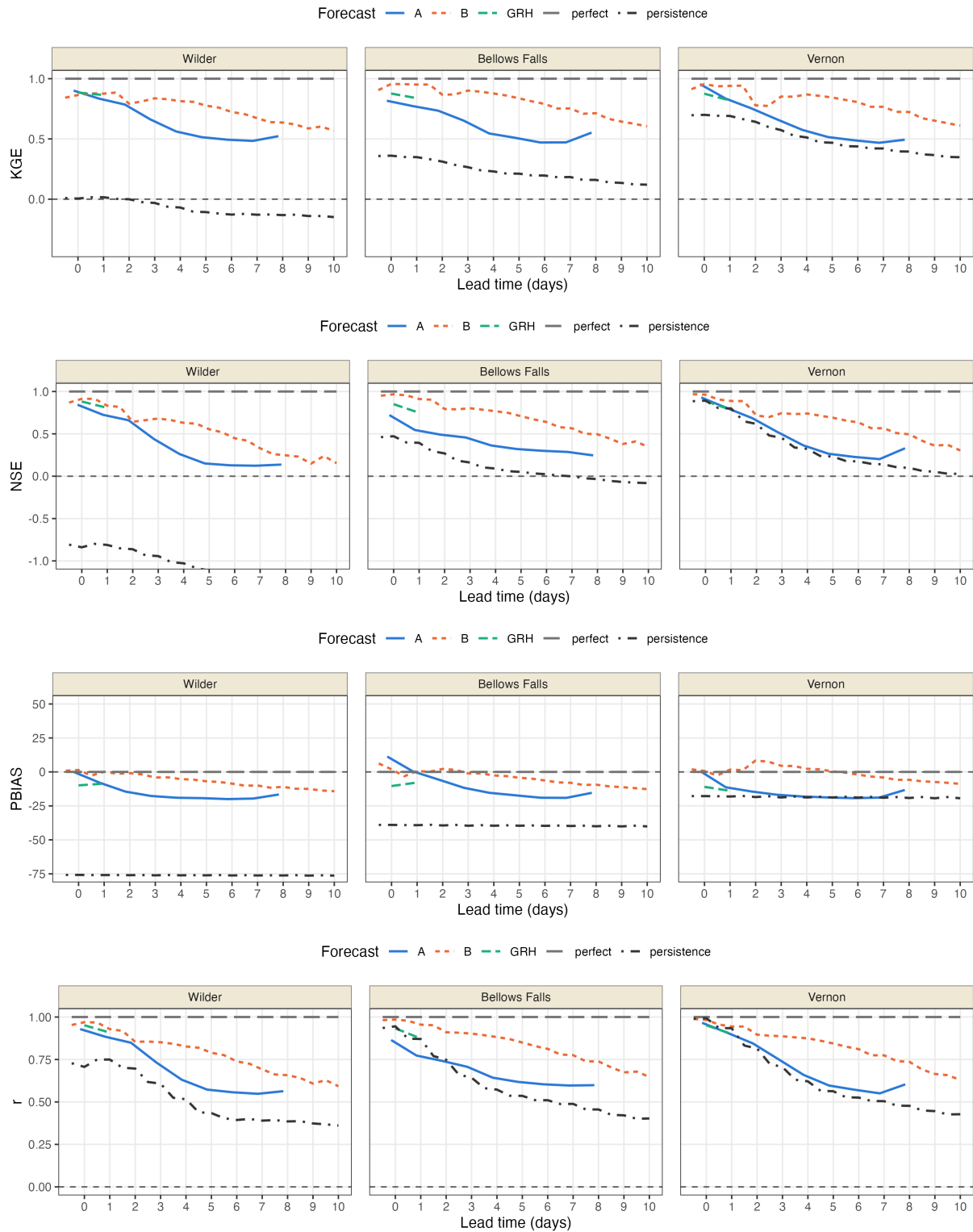


FIGURE 2 | Forecast performance metrics by lead time. A and B are commercial forecasts, GRH is an in-house forecast. The dashed line indicates the threshold where a forecast has no skill over the mean of the observations, which is NSE of 0 and KGE of -0.41 . The persistence benchmark is a meaningful reference only over the first two days; beyond that it is shown only for demonstration purposes.

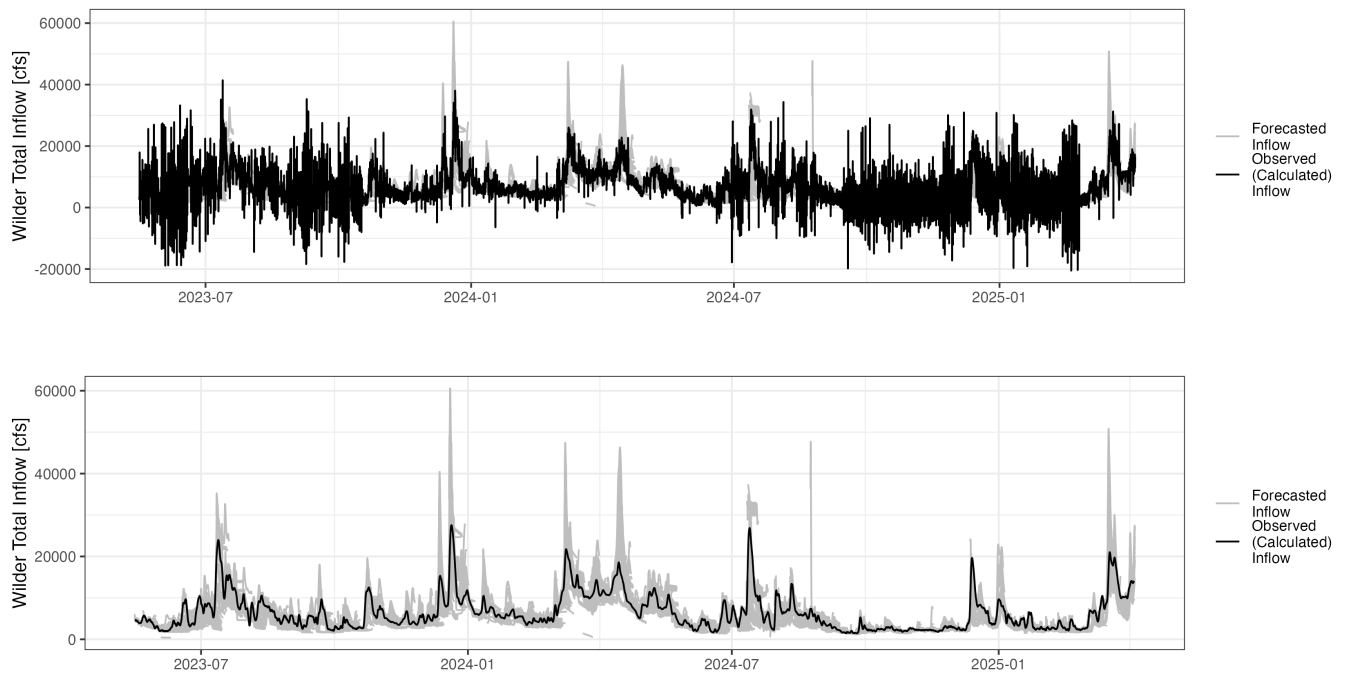


FIGURE 3 | Unsmoothed (top) and smoothed (bottom) calculated inflow at Wilder Dam. Black lines show calculated inflow derived from forebay observations and dam outflow; gray lines show forecasts. Smoothing uses two successive 24-hour moving averages, yielding a triangular kernel spanning 47 hourly observations. Flows are in cubic feet per second (1,000 cfs $\approx$ 28.3 m³/s).

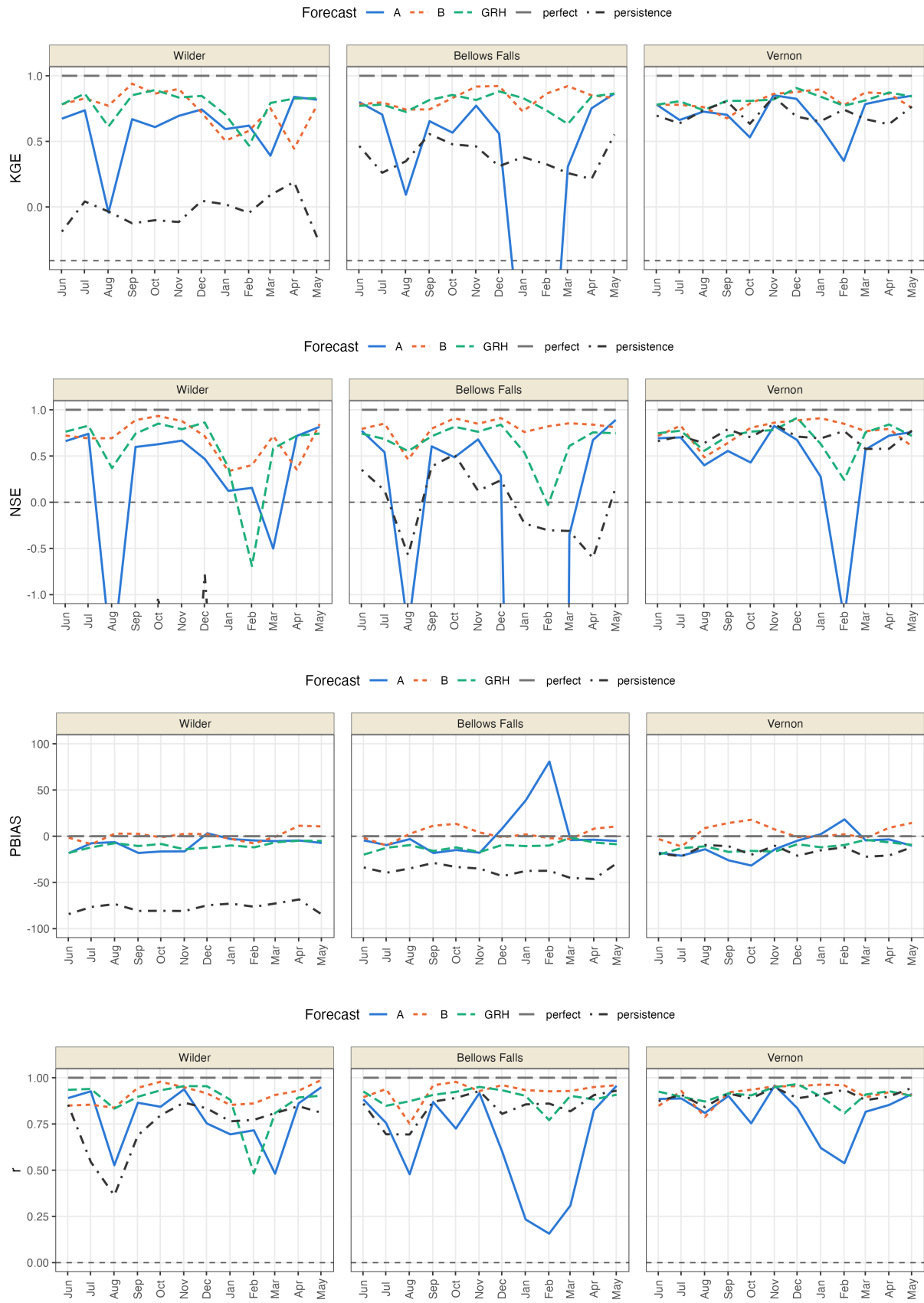


FIGURE 4 | Forecast performance metrics by month. A and B are commercial forecasts, GRH is an in-house forecast. The dashed line indicates the threshold where a forecast has no skill over the mean [of the observations, which is NSE of 0 and KGE of -0.41.](#)

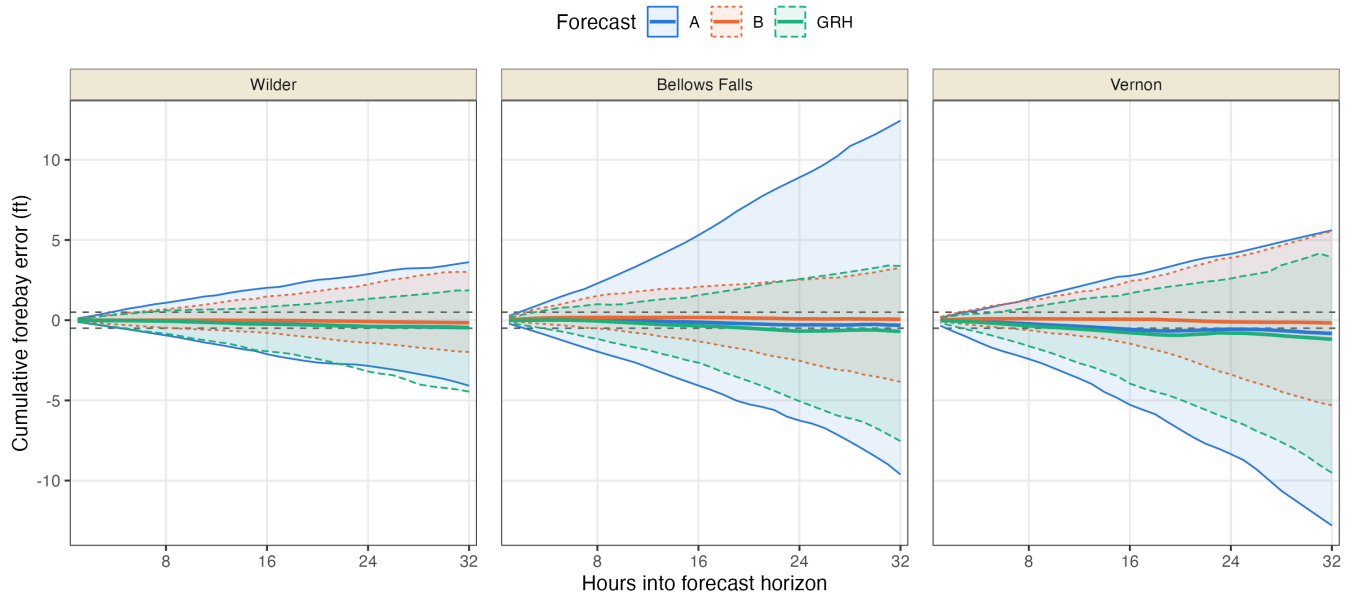


FIGURE 5 | The cumulative probability of exceeding Cumulative forebay error under an idealized pass-inflow stress test with a neutral starting position. Lines show the median across forecast issues and shaded bands span the 2.5th to 97.5th percentiles. Dashed lines mark the ± 0.5 foot (≈ 15 cm) forebay operating range if band. The horizontal axis counts hours from the operations were set start of each product's delivered horizon rather than absolute lead time from issue, so that every point compares an equal number of accumulated hours across products. Forecast A is issued at 04:00 but its delivered series usually begins 25 hours later; its 32 plotted hours therefore span mean lead times of 18.6 to pass inflow (inflow=outflow) where inflow was determined by 49.7 hours, while B and the various inflow forecasts in-house forecast begin at lead hour 1. The window ends at 32 hours, the longest horizon the in-house forecast provides.

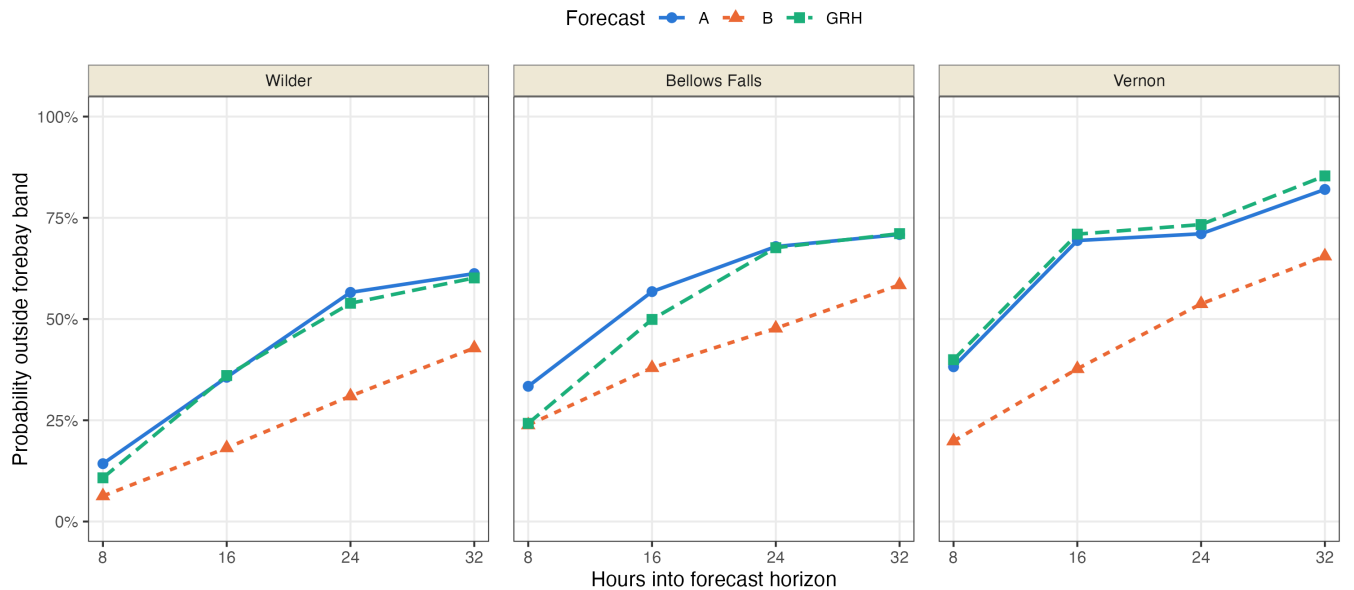


FIGURE 6 | Probability of exceeding a the ± 0.5 foot (≈ 15 cm) forebay operating range given band under the idealized pass-inflow stress test. Each forecast issue begins from a neutral forebay starting position on day 0 (top) and probability. As in Figure 5, the horizontal axis counts hours from the start of exceeding a 0.5-foot operating range on each product's delivered horizon rather than absolute lead time from issue, and the second day window ends at 32 hours, given the forebay starting position from longest horizon the first day (bottom) in-house forecast provides.